

Informatica® RulePoint
6.2

Release Guide

Informatica RulePoint Release Guide
6.2
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Preface

The *Informatica RulePoint 6.1 Release Guide* is for administrators who are responsible for installing, upgrading, and configuring RulePoint. The *Informatica RulePoint Release Guide* lists new features and enhancements, and behavior changes between versions. This guide assumes that you have knowledge of the RulePoint features.

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CHAPTER 1

RulePoint 6.2

This chapter includes the following topics:

- New Features and Enhancements, 7
- Changes to RulePoint 6.2, 8

New Features and Enhancements

RulePoint 6.2 includes the following new features and enhancements:

Single-Node Topology at Installation

The RulePoint installer creates a topology that is simple with a single host and a node. The node is embedded in the grid manager and contains the application services. Application services include a single instance each of the source controller, responder controller, event processor, and activity manager. The topology components communicate with each other through Interprocess Communication - Ultra Messaging (IPC-UM).

After you install RulePoint, you can create a distributed topology by adding more hosts and nodes. Each node is a JVM process and can host one or more types of application and system services. You can configure the services to work in standalone or high availability mode. The system services are the grid manager and the messaging components. Communication between all the components is through the UM store and LBMRD.

High Availability in Activity Manager

You can configure the activity manager to run on one or more backup nodes. When the activity manager becomes unavailable, the backup instance continues to operate on the same node or a backup node.

Additional RulePoint Objects

The predefined list of objects now includes Kinesis and CloudWatch objects.

Amazon Kinesis

You can configure a Kinesis source, responder, and response in RulePoint. You can use the Amazon Web Services connection to connect to Amazon Kinesis. The Amazon Kinesis source reads messages from Amazon Kinesis, which is a managed service for real-time streaming of big data. The Amazon Kinesis responder responds to rule activations by writing a message to the Kinesis stream.

Amazon CloudWatch

You can configure a CloudWatch source, responder, and response in RulePoint. You can use the Amazon Web Services connection to connect to Amazon CloudWatch. The CloudWatch source reads the collected metrics from Amazon CloudWatch and publishes the metrics as events in RulePoint. You can

monitor the metrics in CloudWatch and publish custom metrics to CloudWatch by using the CloudWatch responder.

Amazon Machine Image

The Amazon Machine Image (AMI) for RulePoint contains the Linux instance with RulePoint preinstalled on EC2. You can use the Amazon Web Services Management Console to start a RulePoint instance. After the RulePoint instance starts, log in to the RulePoint Console. For instructions to use the RulePoint AMI, see the *RulePoint Quick Start Guide for Amazon Web Services*.

Support for SOAP 1.2

This release provides support for SOAP 1.2 version for web services.

Changes to RulePoint 6.2

The following changes apply to RulePoint in version 6.2:

Starting RulePoint

Effective in version 6.2, in a single-node topology, you must start only the topology and design-time instance. You need to start the host agent only when you create a multinode topology.

Previously, you had to start the host agent, topology, and design-time instance even for a single-node topology.

Event Set Behavior

Effective in version 6.2, when selecting topics, you can specify the removePrevious event set behavior, which removes all previous events and keeps only the latest event for future evaluation.

Previously, you could specify only removeAll, removeOldest, and removeNone to dispose of events after they participated in rule activation.

CHAPTER 2

RulePoint 6.1

This chapter includes the following topics:

- New Features and Enhancements, 9
- Changes to RulePoint 6.1, 10

New Features and Enhancements

RulePoint 6.1 includes the following new features and enhancements:

Integration with Ultra Messaging Applications such as PowerExchange for Ultra Messaging and Vibe Data Stream for Machine Data

You can configure an Ultra Messaging source, responder, and connection in RulePoint to connect to an external Ultra Messaging source and target. You can also configure RulePoint to connect to Ultra Messaging products in Informatica such as PowerExchange for Ultra Messaging and Vibe Data Stream for Machine Data.

Integrated Event Viewer

You can view a summary of all events that flow into the application in the event viewer. You can view all events or search for strings across events. You can also export events to a .csv file.

Integrated Log Viewer

You can view the details of the run-time logs for the node and the application services on the dashboard. You can filter logs generated for the previous day, previous week, or for a time period that you specify.

Hash, Date, and String Analytics

The predefined analytics list now includes the hash, date, and string analytics.

Rule Profiling

The rule profiler helps you trace the activation of the DRQL statement. The trace lets you know what conditions have passed the evaluation. You can view the evaluation status for each condition in the DRQL statement.

Command Line Programs for Administration Tasks

You can use the command line programs to administer the RulePoint topology and objects. For example, you can import and export objects, perform deployment-related tasks, and encrypt or decrypt passwords.

DRQL Visual Editor and Samples

You can enter multiline comments in the DRQL editor. You can use the DRQL samples available in the installation folder to create advanced rules or as a reference when you create advanced rules.

Save and Update Secondary Objects

You can save and update secondary objects in the run-time environment. You can redeploy primary objects from the associated secondary objects.

User Interface

The following views and interaction of the user interface are enhanced:

- The topic filtering window displays on the main screen of the **Design** tab.
- The **Details View** displays the objects on the left side and the details for the selected objects on the right side.
- A menu of actions displays inline for each selected object.
- The action items on the menu contain the icon and the menu text for the actions.
- The dashboard displays a visual representation of the topology, along with the configured host, nodes, application and system services.

Event and Trace Preview

When you deploy a source, you can use the event preview to view the events for that source. You can also preview the rule trace for a deployed rule.

H2 Database

You can use the embedded open source H2 database bundled with the installer. The option appears if you choose the typical installation mode.

Enhanced Engine Performance

The event processor shows improved performance when compared to version 6.0.1.

Samples

The installation package contains the following samples:

- Banking, Currency Trading, and Business Process Management samples. Contains an SQL script, cust_acc.sql, to create sample tables in the database that you want to use as an event source. Contains documents for each use case with instructions to start RulePoint, create a sample project for the respective use case, and create and deploy sample RulePoint objects for the project. An XML file contains all the RulePoint objects that are created for the use case. If you do not want to manually create the RulePoint objects, you can quickly create the sample project and objects by uploading and importing the XML file to RulePoint. You also have custom services jars if you do not want to manually create the jars for custom services.
- SDK samples. Contains the source code and binaries for samples that you can build and deploy. Custom services include sources, connections, analytics, responders, and marshallers.
- Authentication Plugin samples. Contains the source code and binaries that you can use to build and enable a custom authentication filter.
- Java Adapter samples. Contains the Java samples for REST adapter APIs.

Changes to RulePoint 6.1

The following changes apply to RulePoint in version 6.1:

Export Secondary Objects

Effective in version 6.1, you can export secondary objects with or without dependent objects.

Previously, you could export only primary objects. Only supporting objects referenced in the primary objects were exported.

Support for Multiple RTAM and Watchlist Responders

Effective in 6.1, you can deploy multiple RTAM responders and watchlists from different projects.

Previously, you could deploy only one RTAM responder from a project.

Access Control Lists (ACLs) and User Management

Effective in 6.1, you can associate ACLs to a role. You can also transfer the ACLs to a local or LDAP user or role, before you delete the user or role. You can also force delete users or roles without transferring the ACLs.

Previously, you could not associate the ACLs to role, or transfer the ACLs. You had to first delete the ACLs and then delete the users.

Windows Services to Start the RulePoint Instances

Effective in 6.1, you can use Windows services to start the RulePoint design time, host agent, and topology instances.

Previously, you could use only the command prompt to start the RulePoint instances.

CHAPTER 3

RulePoint 6.0.1

This chapter includes the following topic:

- New Features and Enhancements , 12

New Features and Enhancements

RulePoint 6.0.1 includes the following new features and enhancements:

- RulePoint supports Microsoft SQL Server.
- You can upgrade from RulePoint 6.0 to 6.0.1.
- The environmental variable YOSEMITE_HOME is changed to RULEPOINT_HOME. You need to set the RULEPOINT_HOME environmental variable accordingly.
- The dashboard includes events generated by an event transformer response for the destination topic. The dashboard also displays errors related to rules and analytics.
- You can create object names that contain dots (.).
- RTAM responder includes the "Send Alert" functionality.
- You can use the Java Adapter for REST API to create applications in Java that call the REST APIs.
- You can use the *Developer Guide* for information on custom services, REST APIs, and Java Adapter for REST APIs.

CHAPTER 4

RulePoint 6.0

This chapter includes the following topics:

- New Features and Enhancements, 13
- Changes to RulePoint 6.0, 15

New Features and Enhancements

RulePoint 6.0 includes the following new features and enhancements. This list does not include features introduced in the previous release.

User Interface

RulePoint 6.0 has the following enhancements for all development and administration activities in the user interface:

- Role-specific segmentation of views for developers, administrators, and other user-defined roles
- Projects support to build and manage multiple applications in a single design repository
- Single-page interaction model for listing objects, navigation, search, and specific actions by object type
- Workspace views for multi-object authoring for sources, rules, and responders
- Ability to view all project objects while authoring supporting object types
- Graphical view of related objects across object types
- Bulk creation of topic properties and bulk copy of objects
- Filtering of objects by all related objects
- Dynamic loading of custom sources, analytics, and responders into the design-time and run-time environment using OSGi
- Integrated run-time topology management to perform the following functions:
  - Define and manage the topology
  - Create and manage physical hosts and logical nodes in the topology
  - Create and manage event processing services in the topology
  - Define and manage configuration of services for high availability
  - Configure parameters for all services through the user interface

- Project deployment to the run-time topology and deployment management. You can perform the following deployment-related activities:
  - Partition objects in a project across different run-time services
  - Dynamically deploy, undeploy, and redeploy objects to event processing services
  - Dynamically re-assign objects from one service to another
  - Define deployment policies for rule templates
- Integrated run-time dashboard to perform the following functions:
  - View run-time services and their operational state information
  - View resource utilization, such as CPU and memory of the host where the services run
  - View objects deployed in each run-time service instance and the ability to start and stop
  - View object run-time activity using time series views
  - View all incoming events, outgoing responses, and their payload
  - View activity metrics and state for all deployed objects
  - View execution errors generated by deployed objects
  - Purge statistics at object, service, and topology level
  - Perform dashboard functions based on role-based access
- Integrated Real-Time Alert Manager
- Sample project with all object types
- Sample custom service sources

Programming Model

RulePoint 6.0 has the following features and enhancements in the programming model:

- New object type called connections
- Dynamic schedules for all sources
- Pluggable marshallers to enrich events
- Pluggable unmarshallers to enrich responses
- Additional geospatial, math, and string analytics, which add to 57 built-in analytics
- Ability to configure rules to run concurrently for out-of-order processing of events
- New object lifecycle states to support change management of objects
- Detect and Respond Query Language (DRQL) enhancements:
  - “group by” selector operator replaces “match()” analytic
  - “distinct” selector operator replaces “unique()” analytic
  - “remove” post-processing operator replaces “removeall()”, “removenone()”, “removeoldest()” analytics
  - Sliding windows by time or event count
  - Tumbling windows by time or event count

Scalability, Performance, and Reliability

RulePoint 6.0 has the following features and enhancements:

- Ability to build a run-time topology that consists of N hosts. Each host can contain N nodes, each node can have N event processing services, and each service can have N user-defined application objects
- Source controller service to vertically and horizontally scale, partition, and isolate source executions
- Event processor service to vertically and horizontally scale, partition, and isolate rule executions
- Responder controller service to vertically and horizontally scale, partition, and isolate responder executions
- Activity manager service to collect and store statistics of all objects across the topology
- Execution infrastructure, called as node, to host one or more event processing services
- Host agent service to collect resource utilization information and manage the node lifecycle on hosts
- Grid manager service to manage deployment, lifecycle, and resilience of event processing services across a grid of physical hosts and logical nodes
- Ability to run event processing services in standalone mode and high-availability mode
- Guaranteed event delivery and processing data exchange infrastructure by using Ultra Messaging Persistence
- Parallel persistence of all incoming events and activations for recovering states across cold restarts in the topology
- Support for creating multiple stores for guaranteed persistence to guard against disk failures
- Database support for source controller state synchronization between primary and secondary instances
- Ultra Messaging support for event processor state synchronization between primary and secondary instances
- Performance improvement up to 2x from earlier RulePoint releases, on a single-host topology that runs on the same hardware, based on an internal benchmark
- System-managed failover support for all event processing services
- Failure fencing and resiliency of event processing services

Supportability

RulePoint 6.0 has the following features and enhancements:

- Error codes and descriptions for errors in the design-time and run-time environment
- Email alerts for reporting failure in the topology and services
- Command line tool to collect and package topology configuration and logs for customer support analysis

Changes to RulePoint 6.0

The following changes apply to RulePoint in version 6.0:

Object names

Effective in version 6.0, you cannot have spaces, "-", or "." for object names. Informatica recommends you use a combination of business-centric prefixes and object types to name RulePoint objects. For example, for a banking application, you can name the topic as "DebitT_transactions."

Previously, you could have spaces, "- ", or "." for object names.

Event rate and last modified date display

Effective in version 6.0, you can use the **Dashboard** view to see the run-time statistics for RulePoint objects.

Previously, the autocomplete feature displayed the incoming event rate along with the published date.

Application container support

Effective in version 6.0, RulePoint 6.0 is certified only against Apache Tomcat 7.0.25. Tomcat is also bundled and installed along with the RulePoint installation.

Previously, RulePoint supported application containers, such as WebSphere and WebLogic, in addition to Apache Tomcat.

isBatchComplete() analytic

Effective in version 6.0, a predefined system topic called system_markers ensures that an event is generated when a batch completes processing. You can write rules to listen to the marker topic for a batch of events.

Previously, rules use isBatchComplete() analytic to check if a specific batch has completed processing.

Filtering

Effective in version 6.0, you can filter objects by related objects. For example, you can find rules by analytics or watchlists, or find sources by topics. You can also organize application objects as projects.

Previously, a tree structure reflected the organization of topics. You could filter only specific objects by selecting the topics in the tree structure.

Status of object types

Effective in version 6.0, RulePoint has a distinct design-time and run-time environment. You must deploy objects to the application services to run sources, rules, and other object types. The state of objects are defined based on the deployment status. The "Active" state now maps to "Deployed" state and "Disabled" state maps to "Undeployed" state.

Previously, object types displayed as active, favorite, and disabled status.

Audit record listener

Effective in version 6.0, RulePoint does not support audit record listener.

Previously, the audit record source service listens for notifications when a transaction is logged in the RulePoint audit record topic and publishes an event for the transaction.

Batch completion listener

Effective in version 6.0, you can use system markers, which ensure that an event is generated when a batch completes processing. You can write rules to listen to this topic, so that they get executed when the batch is complete.

Previously, batch completion listeners were used to indicate that the processing of a set of events is complete.

Rule reviews

Effective in version 6.0, RulePoint has a distinct design-time and run-time environment, which ensures that only users with specific permissions can deploy objects to the run-time environment. The state of the modified object changes to "Needs_Deployment," so that administrators can view the modified objects, review, and deploy them.

Previously, an administrator had to review the rule or template a user creates or modifies before it becomes active.

Event Publisher

Effective in version 6.0, RulePoint natively supports multiple instances of application services that exchange data over a grid of hosts.

Previously, event publisher instances were used that send events to other RulePoint instances.

Event Subscriber

Effective in version 6.0, RulePoint natively supports multiple instances of application services that exchange data over a grid of hosts.

Previously, event subscriber instances were used for receiving events from other RulePoint instances.

Event Recorder

Effective in version 6.0, RulePoint supports recording of events and responses across topics and responses through a simple, configuration driven-approach.

Previously, event recorder instances were used to persist events.

Watchlist-based picklist

Effective in version 6.0, you must use a picklist of topics for the parameter topic_name in the DRQL, for example, WHEN 1 <<topic_name>> t0 THEN "RTAM Response".

Previously, you used a watchlist-based picklist for parameterized topics in a template.

Watchlist keyword

Effective in version 6.0, when you create a template that uses a watchlist, do not use the keyword watchlist before the watchlist name. For example, you must rewrite `isHoliday(tran_dt,watchlist:<<holiday>>))` as `isHoliday(tran_dt, <<holiday>>)`.

Previously, you had to add the watchlist keyword when using a watchlist in a template, for example, `isHoliday(tran_dt,watchlist:<<holiday>>))`.

Query for SQL source

Effective in version 6.0, when you add a query in the "update" field of the SQL source, that query runs only when the SQL source completes fetching all the records.

Previously, when you add a query in the "update" field of the SQL source, that query runs for every record fetched by the SQL source.